

5.3 MOSFET Circuit Models

5.3.1 MOSFET DC Model

- Square-Law Theory: Model Equations
- MOSFET Bias Circuits
- Q-Point Analysis

5.3.2 MOSFET Small Signal Model

- Y-Parameter Two-Port Network
- Small Signal Equivalent Circuit
- Small Signal Analysis

5.3.3 Comparison MOSFET – BJT

5.3.4 Common-Source Amplifier

Literature: Jaeger, Blalock, Chapter 4.6-4.9, page 167-187
 Jaeger, Blalock, Chapter 13.8-13.10, page 815-838
 Jaeger, Blalock, Chapter 15.4, page 1016-1024
 Jaeger, Blalock, Chapter 16.2, page 1049-1063

5.3.1 DC Models

– NMOS & Square-Law Theory –

	Long Channel Approximation	Short Channel Approximation
Saturation Region $V_{GS} > V_T; V_{DS} > V_{Dsat}$	$I_D = \frac{K_n}{2} (V_{GS} - V_T)^2$	$I_D = \frac{K_n}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$
Triode Region $V_{GS} > V_T; V_{DS} \leq V_{Dsat}$	$I_D = K_n \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$	$I_D = K_n \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$
Cut-Off Region $V_{GS} < V_T$	$I_D = 0$	$I_D = 0$

$$K_n = \mu_n C_{ox} (W / L)$$

In addition, we assume $I_G = 0$ and $I_B = 0$ (substrate current)

How to Perform Q-Point Analysis?

1. Educated guess: Assume the region of operation (most often the saturation region)
2. Use circuit analysis (might require Thévenin equivalent circuit) to find V_{GS}
3. Use V_{GS} to calculate I_D , and I_D to determine V_{DS} (using MOSFET model and KVL)
4. Check the validity of the operation region assumptions
5. Change assumptions and analyze again if necessary

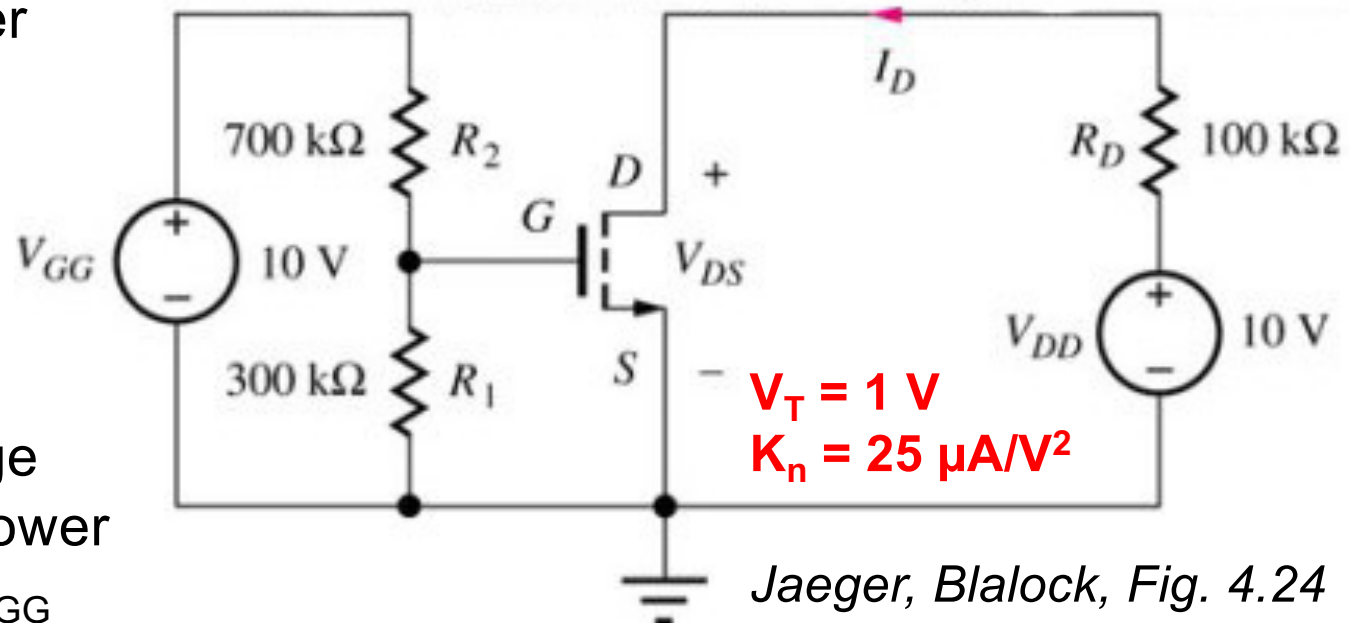
Constant Gate-Source Voltage Bias

- Voltage divider biases gate terminal
- With $I_G = 0$, we find $V_{GS} = 3 \text{ V}$
- $R_1 + R_2$ is large to minimize power supplied by V_{GG}
- Assuming biasing in the saturation region, we find

$$\text{Saturation Model: } I_D = \frac{K_n}{2} (V_{GS} - V_T)^2 = 50 \mu\text{A}$$

$$\text{KVL: } V_{DD} = I_D R_D + V_{DS} \Rightarrow V_{DS} = 5 \text{ V}$$

- Because $V_{DS} = 5 \text{ V} > V_{GS} - V_T = 2 \text{ V}$, the MOSFET is in saturation



Constant Gate-Source Voltage Bias (cont.)

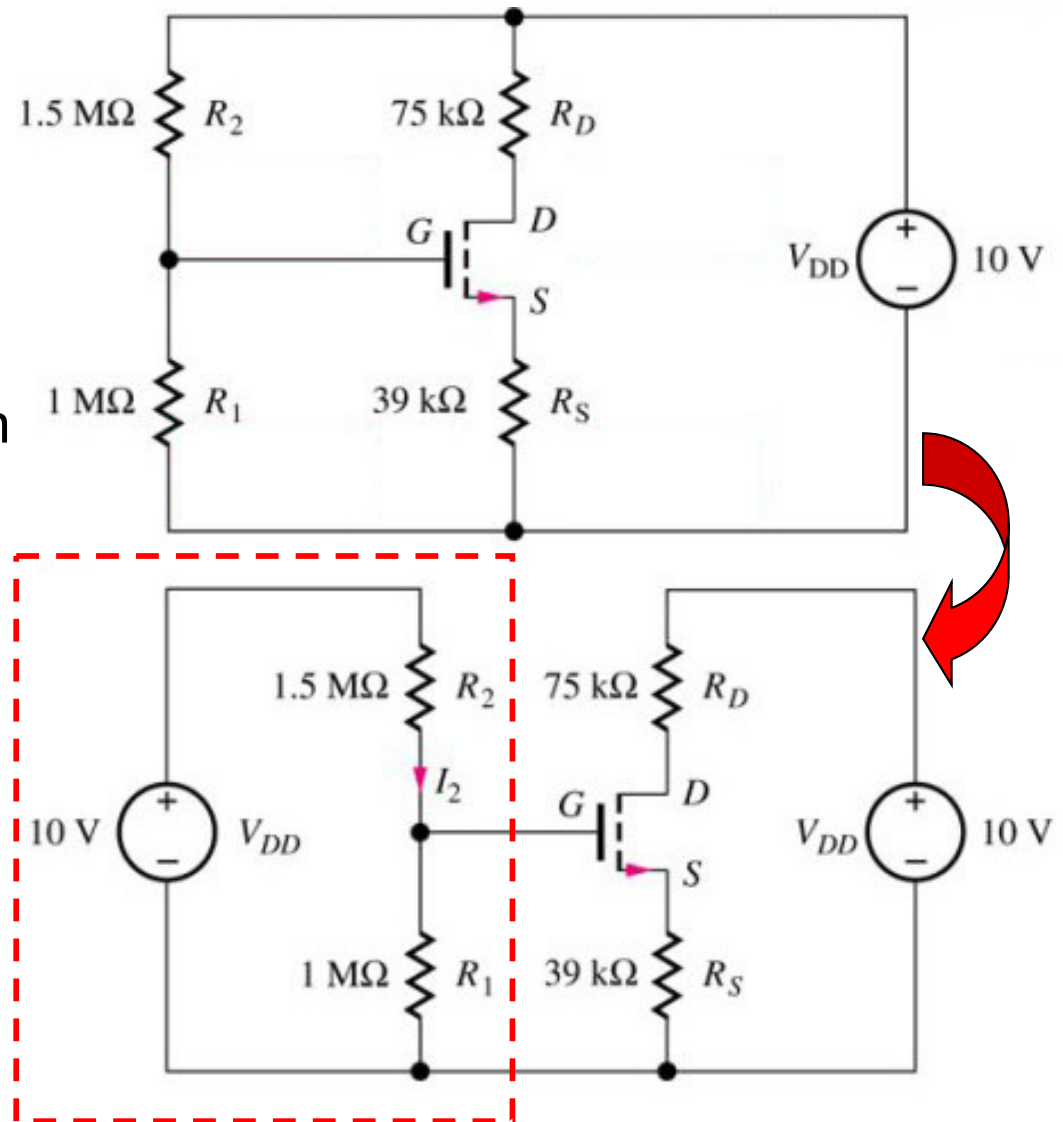
- Q-point is $(I_D, V_{DS}) = (50 \mu\text{A}, 5 \text{ V})$ at $V_{GS} = 3 \text{ V}$
- Note: If we know the mode of operation (in this case saturation), we can calculate V_{GS} from I_D and V_{DS} ; thus, the Q-point is defined by I_D and V_{DS} only
- Why is the constant gate-source voltage bias typically NOT used in practical applications?
 - Obtained Q-point is very sensitive to variations of the transistor parameters (K_n, V_T): I_D increases linearly with K_n and quadratically on V_T
 - “Better” bias circuits: two or **four-resistor** bias, bias with current mirror
- How would you obtain the above Q-point if $\lambda = 0.02 \text{ V}^{-1} \neq 0$?
- How would you perform a load-line analysis of the circuit?

MOSFET Four-Resistor Bias

- Most general and important bias method
- Advantages:
 - Feedback circuit helps stabilize the Q-point upon variation of transistor parameters and temperature
 - Only single voltage supply needed
- Let's analyze this circuit..... $V_T = 1\text{ V}$

$$K_n = 25\text{ }\mu\text{A/V}^2$$

Jaeger, Blalock, Fig. 4.27



MOSFET Four-Resistor Bias

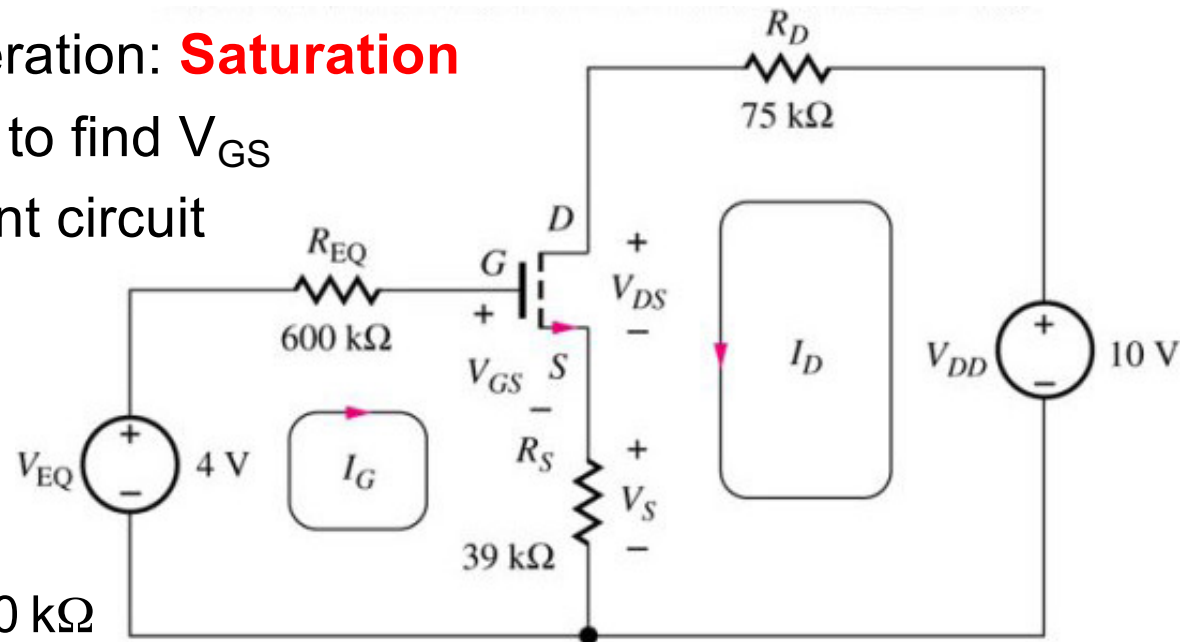
– Analysis –

1. Assume a region of operation: **Saturation**
2. Perform circuit analysis to find V_{GS}

(a) Find Thevenin equivalent circuit for gate bias network (see previous page):

$$V_{EQ} = \frac{R_1}{R_1 + R_2} V_{DD} = 4 \text{ V}$$

$$R_{EQ} = \frac{V_{EQ}}{V_{DD}/R_2} = \frac{R_1 R_2}{R_1 + R_2} = 600 \text{ k}\Omega$$



Jaeger, Blalock, Fig. 4.28

(b) Use KVL and MOSFET model to extract V_{GS}

$$V_{EQ} = \underbrace{I_G R_{EQ}}_{=0, \text{ because } I_G = 0} + V_{GS} + \underbrace{(I_D + I_G) R_S}_{=I_D, \text{ because } I_G = 0} \Rightarrow V_{EQ} = V_{GS} + I_D R_S$$

MOSFET Four-Resistor Bias

– Analysis (cont.) –

Solve quadratic equation to obtain V_{GS}

$$\left. \begin{array}{l} I_D = \frac{K_n}{2} (V_{GS} - V_T)^2 \\ V_{EQ} = V_{GS} + I_D R_S \end{array} \right\} V_{EQ} = V_{GS} + \frac{K_n R_S}{2} (V_{GS} - V_T)^2$$

yielding $V_{GS} = -2.71 \text{ V}$ (which would mean cut-off mode) and
 $V_{GS} = 2.66 \text{ V}$

3. Use V_{GS} to calculate I_D , and I_D to determine V_{DS}

$$I_D = \frac{K_n}{2} (V_{GS} - V_T)^2 = 34.4 \mu\text{A}$$

$$V_{DD} = I_D R_D + V_{DS} + \underbrace{(I_D + I_G)}_{=I_D, \text{ because } I_G=0} R_S \Rightarrow V_{DS} = V_{DD} - I_D (R_D + R_S) = 6.08 \text{ V}$$

4. Check the validity of the operation region assumptions

$$V_{GS} - V_T = 1.66 \text{ V} < V_{DS} = 6.08 \text{ V} \Rightarrow \text{Saturation} \quad \text{😊}$$

MOSFET Four-Resistor Bias

– Analysis (cont.) –

- Resulting Q-point: $(I_D, V_{DS}) = (34.4 \mu\text{A}, 6.08 \text{ V})$
- Why is the bias point “more” stabilized compared to constant V_{GS} bias?
 - Bias circuit uses **negative feedback** to stabilize bias point

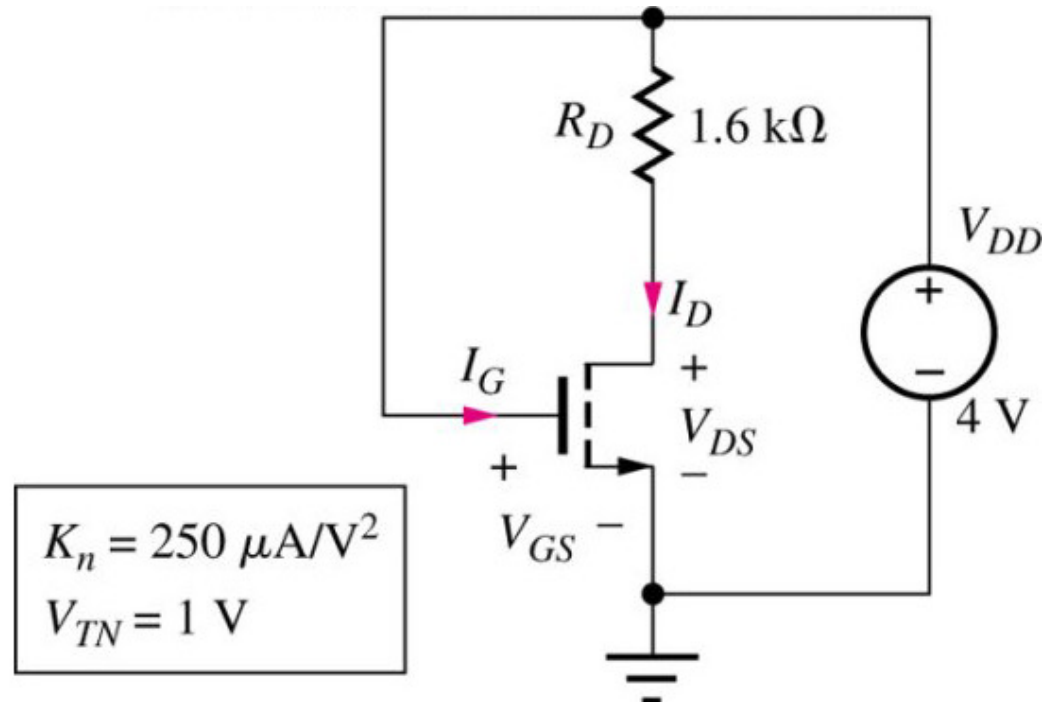
$$I_D = \frac{K_n}{2} (V_{GS} - V_T)^2 \quad \text{and} \quad V_{EQ} = V_{GS} + I_D R_S$$

- If I_D rises (because of parameter or temperature variations), V_{GS} must decrease because V_{EQ} is fixed; but a decrease in V_{GS} will tend to restore (decrease) I_D , thus stabilizing the operating point of the n-channel MOSFET
- In the previous circuits it was assumed that the **bulk contact is connected to the grounded source contact** (three-terminal device); if this is not the case, the analysis become more involved (body effect)

Sample Problem

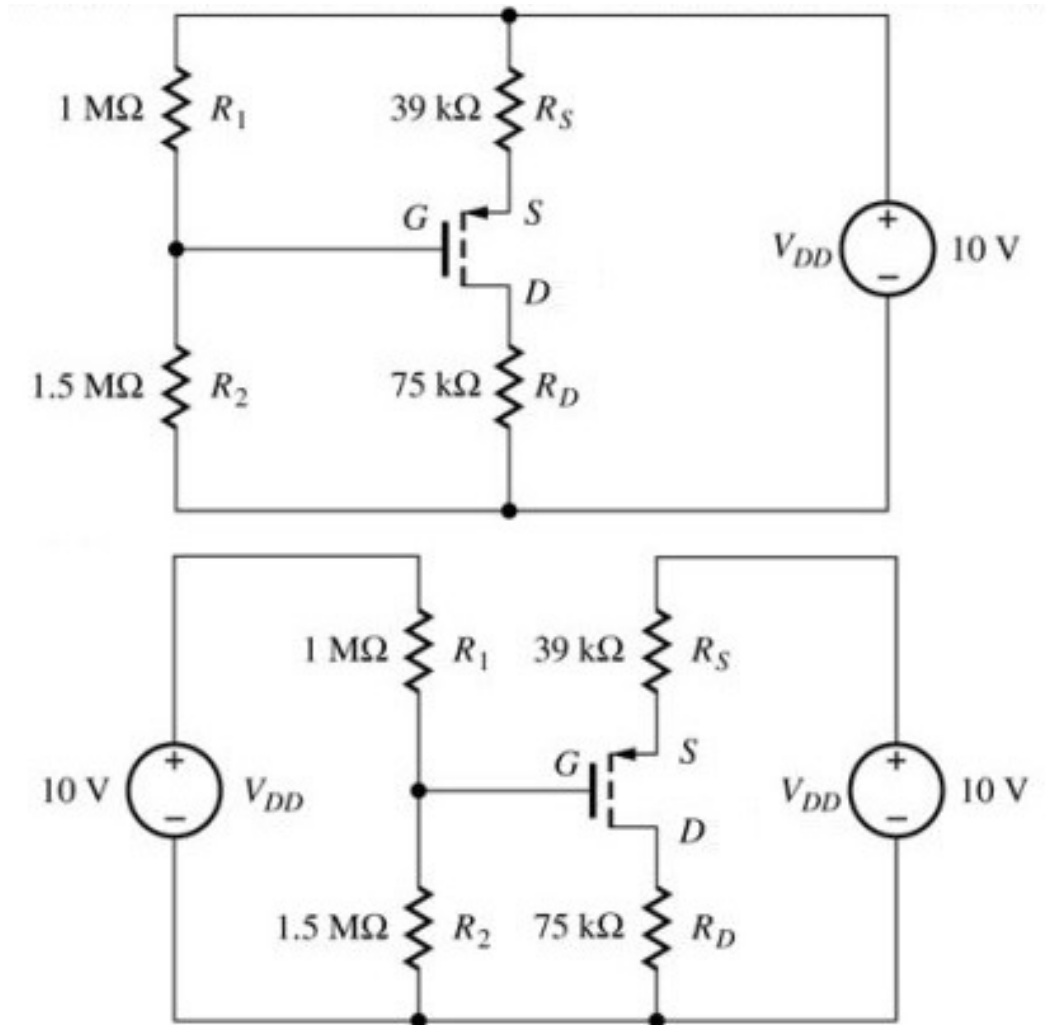
– It's YOUR Turn..... –

- Assumption for Mode of Operation?
- Calculation:



And what about PMOS FETs?

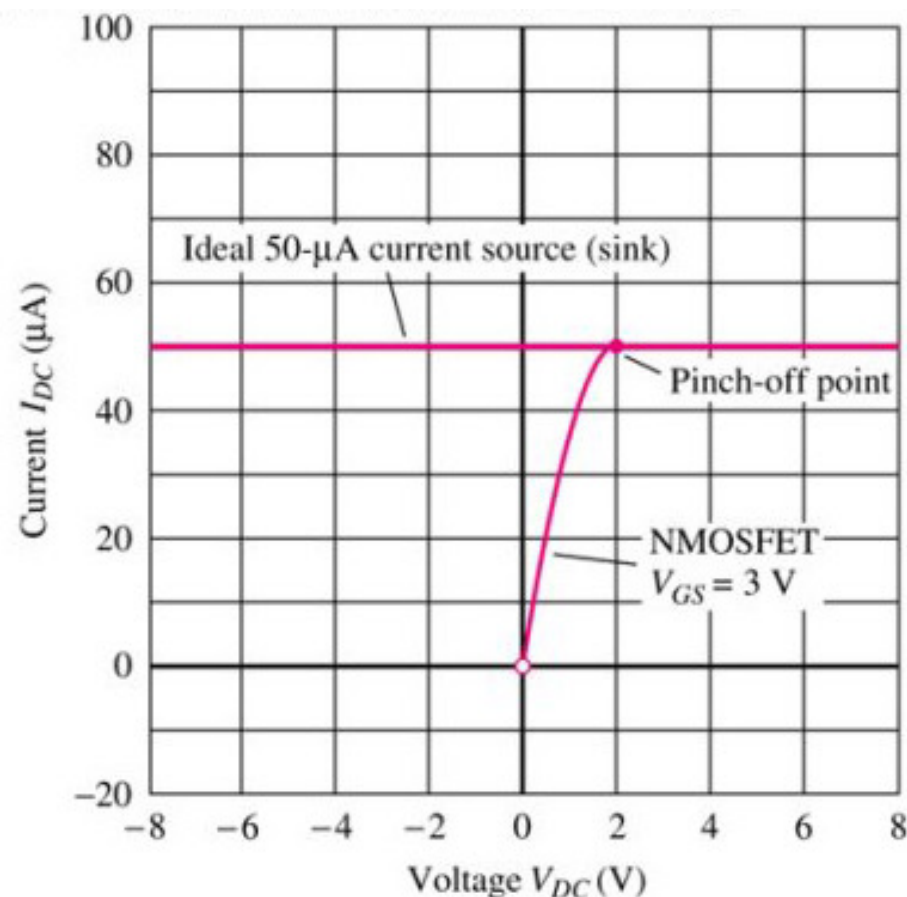
- PMOS bias techniques mirror those of NMOS bias examples
- Source of PMOS transistor is drawn on top of device because it is normally connected to a potential higher than the drain
- For PMOS transistors
 - the drain current I_D is **POSITIVE** when coming **OUT** of the drain terminal and
 - the values for V_{GS} and V_{DS} will be **NEGATIVE**
- Try to analyze the shown PMOSFET bias circuit



Jaeger, Blalock, Fig. 4.30

Current Sources

- Current sources are widely used to **establish transistor working points in integrated circuits**
 - Ideal current source delivers constant current independent of voltage across the source
 - How can we use MOSFETs as effective current sources?
- ➔ MOS transistors operated in saturation region (pinched-off region) deliver constant current for $V_{DS} > V_{Dsat}$



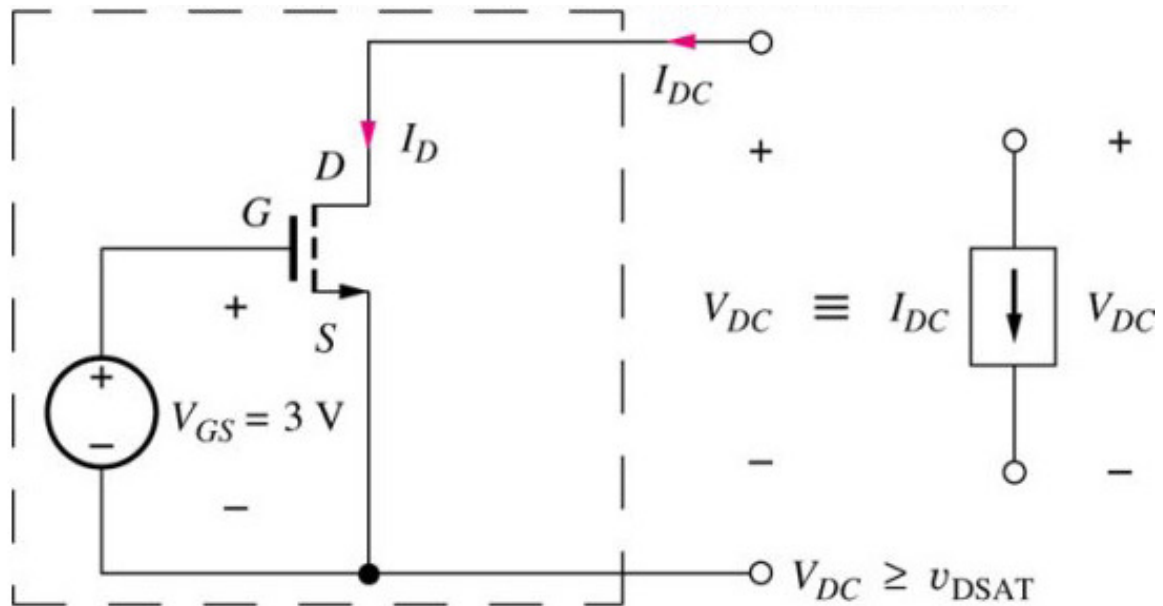
MOSFET Current Source

How can we achieve output characteristic shown on previous page?

- Use NMOS transistor operated with constant gate-source voltage $V_{GS} = 3\text{ V}$ (with $V_{GS} > V_T$)
- As long as $V_{DC} \geq V_{Dsat}$, the MOSFET is biased in the pinched-off region and the drain current I_D remains constant

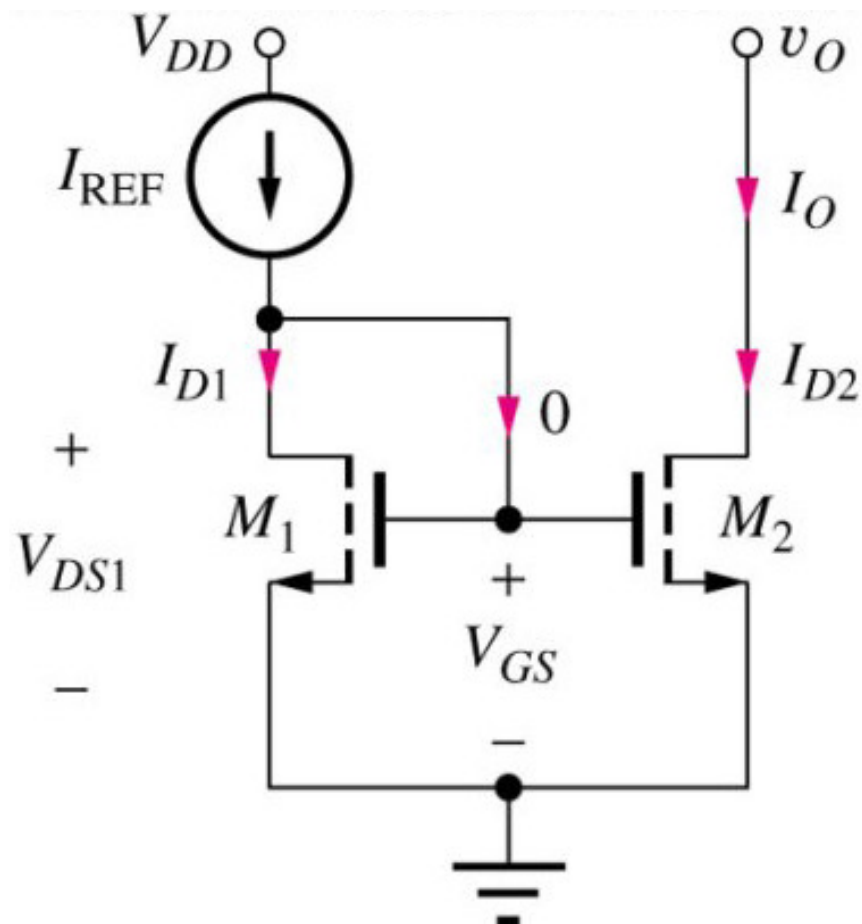
$$I_D = \frac{K_n}{2} (V_{GS} - V_T)^2$$

- Assuming $V_T = 1\text{ V}$ and $K_n = 25\text{ }\mu\text{A/V}^2$, we find $I_{DC} = I_D = 50\text{ }\mu\text{A}$



MOSFET Current Mirror

- Fundamental building block in electronic circuit design defining currents and current ratios (from a single reference current)
- Current mirror relies on **matched transistor parameters V_T , K_n , λ and W/L** (rather than absolute parameter values) to deliver $I_O = I_{REF}$
- Note: in a CMOS process, absolute device parameters might have 20-30% spread, but neighboring transistor parameters are typically matched to within 1-2%; as a result circuit designers rely on ratios rather than absolute values



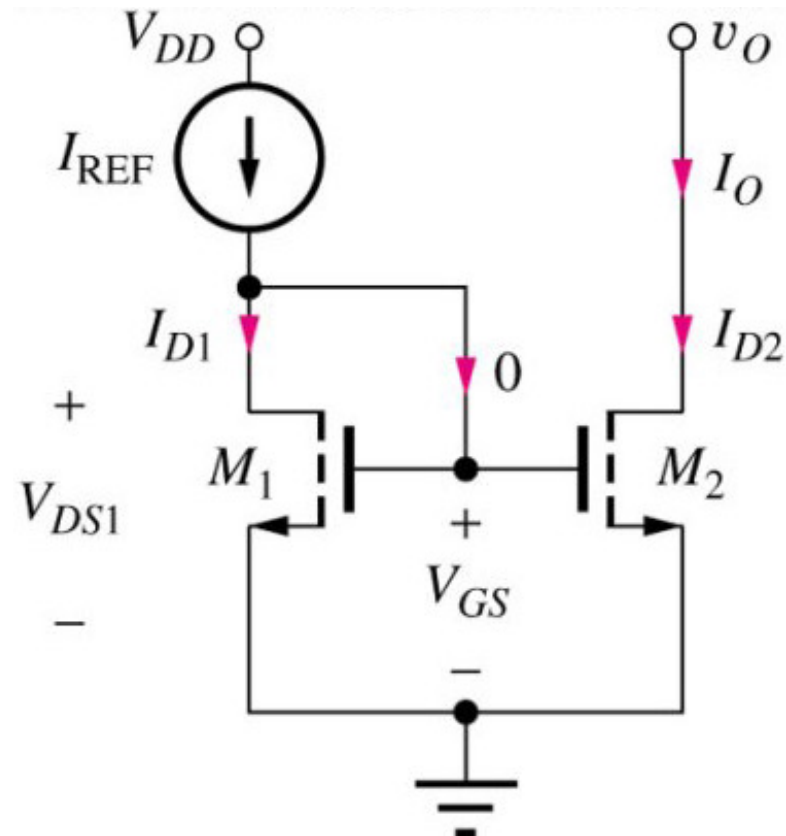
NMOS Current Mirror

- Assumption: Transistors M_1 and M_2 have matched parameters V_T , K_n , λ and W/L
- Gate currents I_G of both transistors are zero, i.e. the reference current must flow into drain of M_1 : $I_{REF} = I_{D1}$
- With $V_{DS1} = V_{GS}$ ($> V_{GS} - V_T$), the transistor M_1 is in saturation and

$$I_{REF} = I_{D1} = \frac{K_n}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS1})$$

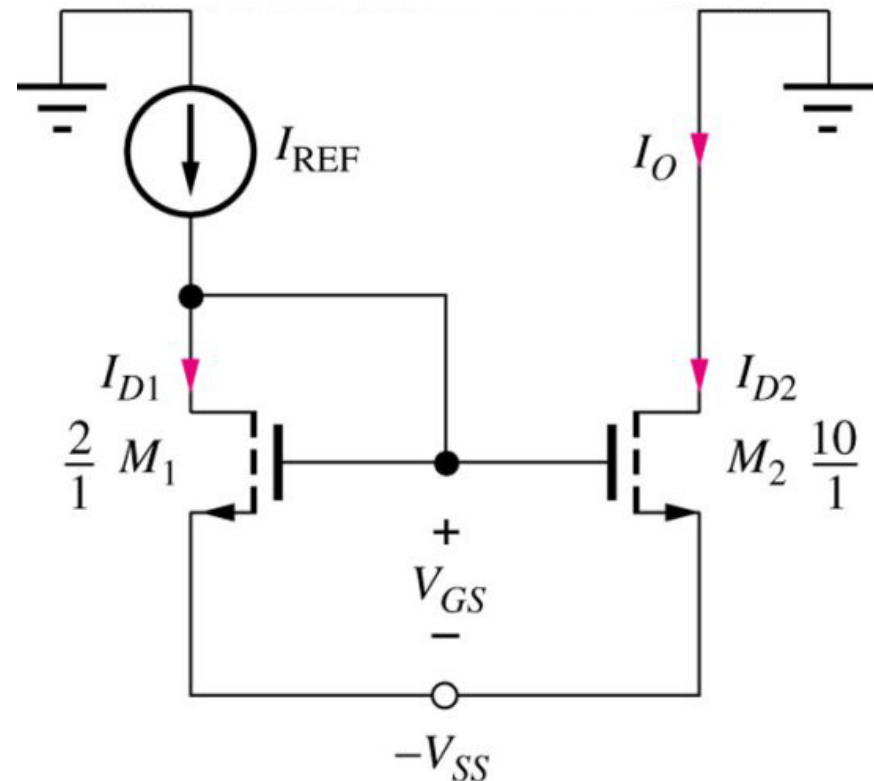
- Similarly, the current through M_2 becomes

$$I_O = I_{D2} = \frac{K_n}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS2}) = I_{REF} \frac{(1 + \lambda V_{DS2})}{(1 + \lambda V_{DS1})} \cong I_{REF}$$



NMOS Current Mirror (cont.)

- For equal values of V_{DS} , the **output current mirrors the reference current**
- Note: In most applications $V_{DS1} \neq V_{DS2}$, resulting in a slight mismatch of the currents
- By designing the transistors M_1 and M_2 with different W/L ratio, current ratios can be defined!



Jaeger, Blalock, Fig. 16.3

$$I_O = I_{REF} \frac{(W/L)_2(1 + \lambda V_{DS2})}{(W/L)_1(1 + \lambda V_{DS1})}$$

Note: Current mirror is operated between ground and negative supply; compared to slide 14 the supply voltage is just shifted

Multiple Current Mirrors

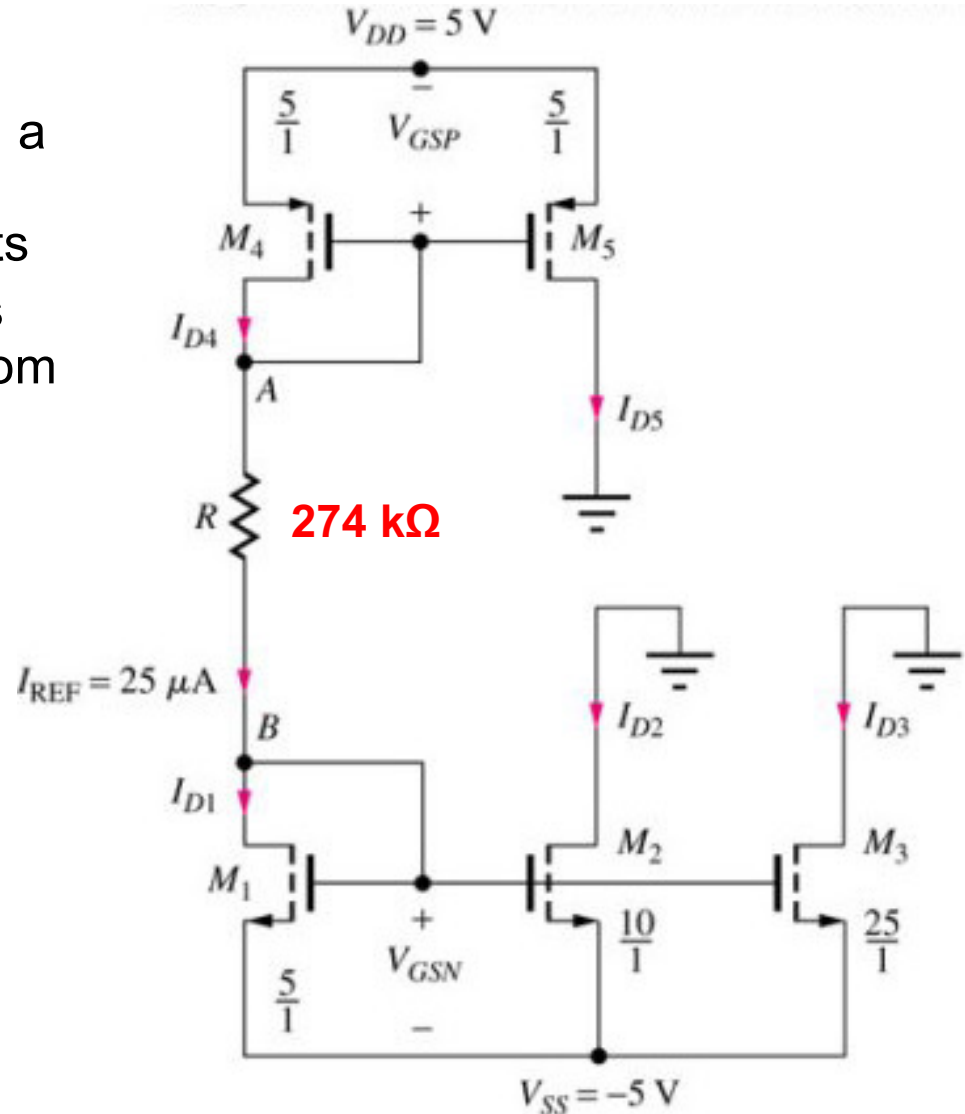
- Current mirrors are used to generate multiple currents from a single reference current, as needed in more complex circuits
- The displayed circuit generates three currents I_{D2} , I_{D3} and I_{D5} from the reference current I_{REF}
- The resistance R defines the reference current with

$$R = \frac{(V_{DD} - V_{SS}) + V_{GSP} - V_{GSN}}{I_{REF}}$$

$$I_{D1} = \frac{K_n}{2} (V_{GSN} - V_{Tn})^2$$

$$I_{D4} = I_{D1} = I_{REF} = \frac{K_p}{2} (V_{GSP} - V_{Tp})^2$$

- Large resistors R are typically realized with a transistor in integrated circuit design



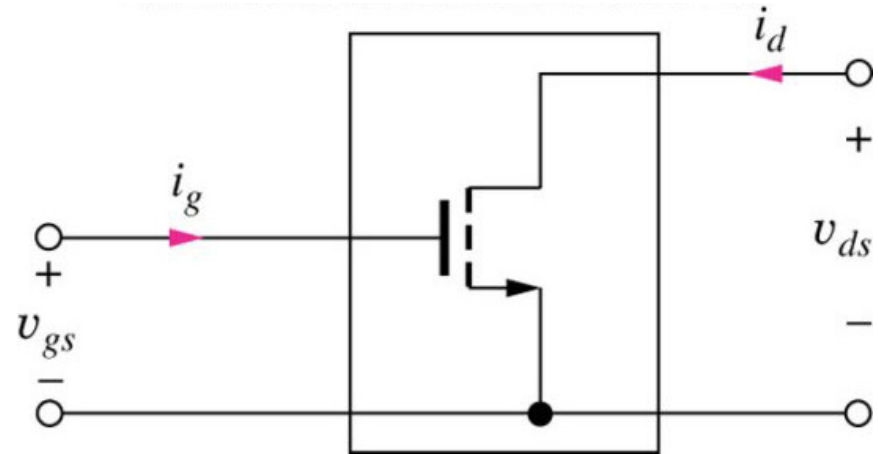
5.3.2 Small Signal Analysis

- Small signal model for FET to analyze (small signal) AC behavior of e.g. common-source amplifier
- DC voltages/currents (all capital letters) superimposed by AC currents/voltage (all small letters)

$$V_{GS} = V_{GS} + v_{gs} \quad V_{DS} = V_{DS} + v_{ds}$$

$$I_G = I_G + i_g \quad I_D = I_D + i_d$$

- Small signal model based on y-parameter two-port network (expressing currents as a function of the voltages)



Jaeger, Blalock, Fig. 13.21



$$i_g = y_{11} v_{gs} + y_{12} v_{ds}$$

$$i_d = y_{21} v_{gs} + y_{22} v_{ds}$$

$$\begin{pmatrix} i_g \\ i_d \end{pmatrix} = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix} \begin{pmatrix} v_{gs} \\ v_{ds} \end{pmatrix}$$

Y-Parameter Definition (cont)

- Y-parameters depend on the DC working point (Q-point) of the MOSFET and have units of $[\Omega^{-1}]$
- Y-parameters are defined as partial derivatives of complete port variables

$y_{11} = \left. \frac{\partial i_G}{\partial v_{GS}} \right _{Q\text{-point}} \approx \left. \frac{\partial I_G}{\partial V_{GS}} \right _{Q\text{-point}}$	$y_{12} = \left. \frac{\partial i_G}{\partial v_{DS}} \right _{Q\text{-point}} \approx \left. \frac{\partial I_G}{\partial V_{DS}} \right _{Q\text{-point}}$
$y_{21} = \left. \frac{\partial i_D}{\partial v_{GS}} \right _{Q\text{-point}} \approx \left. \frac{\partial I_D}{\partial V_{GS}} \right _{Q\text{-point}}$	$y_{22} = \left. \frac{\partial i_D}{\partial v_{DS}} \right _{Q\text{-point}} \approx \left. \frac{\partial I_D}{\partial V_{DS}} \right _{Q\text{-point}}$

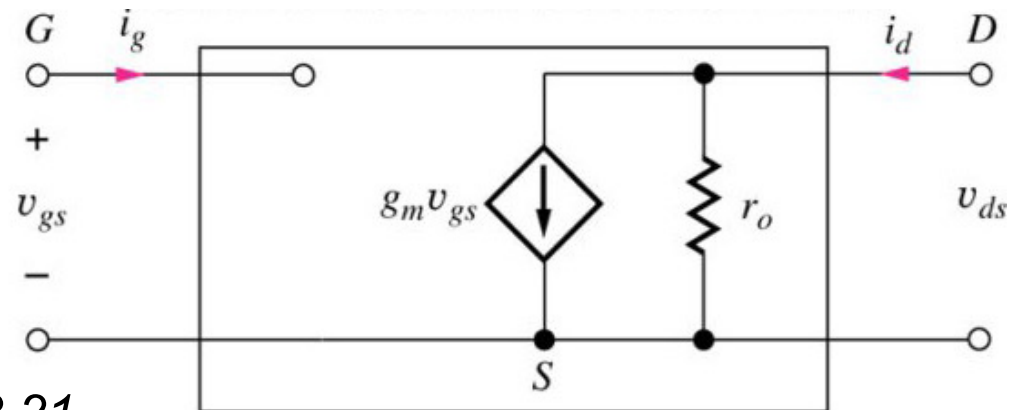
- Derivatives are **evaluated from the large-signal model equations**, e.g. in the saturation region ($v_{DS} \geq v_{GS} - V_T$ and $i_G = 0$)

$$I_D = \frac{K_n}{2} (V_{GS} - V_T)^2 \quad \text{with} \quad K_n = \mu_n C_{ox} (W / L)$$

Y-Parameter Definition (cont.)

– Saturation Mode –

	Long - Channel ($\lambda = 0$)	Short - Channel ($\lambda \neq 0$)	} $i_G = 0$
y_{11}	$= 0$	$= 0$	
y_{12}	$= 0$	$= 0$	
$y_{21} = g_m$	$= K_n (V_{GS} - V_T)$	$= K_n (V_{GS} - V_T)(1 + \lambda V_{DS}) = \frac{2 I_D}{V_{GS} - V_T}$	
$y_{22} = \frac{1}{r_o}$	$= 0$	$= \lambda \frac{K_n}{2} (V_{GS} - V_T)^2 = \frac{\lambda I_D}{1 + \lambda V_{DS}}$	



Jaeger, Blalock, Fig. 13.21

Y-Parameter Definition (cont.)

– Saturation Mode –

- Because the gate terminal is insulated from the channel by the gate oxide, the **input resistance** $1/y_{11}$ of the FET is infinite
- **Transconductance** g_m and **output resistance** r_o :

$$\begin{aligned} g_m = y_{21} &= \frac{I_D}{\frac{V_{GS} - V_T}{2}} \\ &= K_n (V_{GS} - V_T)(1 + \lambda V_{DS}) = \sqrt{2K_n I_D (1 + \lambda V_{DS})} \\ r_o = \frac{1}{y_{22}} &= \frac{1}{\lambda} + \frac{V_{DS}}{I_D} \cong \frac{1}{\lambda I_D} \end{aligned}$$

Limits of Small-Signal Model

- What are the limits of the linear operation of the MOSFET, i.e. what are the limits for the input voltage amplitude v_{gs} to enable use of small-signal model?
- Assume n-channel MOSFET in saturation region with $v_{GS} = V_{GS} + v_{gs}$ and $i_D = I_D + i_d$:

$$i_D = \frac{K_n}{2} (v_{GS} - V_T)^2$$

$$i_D = I_D + i_d = \frac{K_n}{2} (V_{GS} + v_{gs} - V_T)^2 = \frac{K_n}{2} [(V_{GS} - V_T)^2 + 2(V_{GS} - V_T)v_{gs} + v_{gs}^2]$$

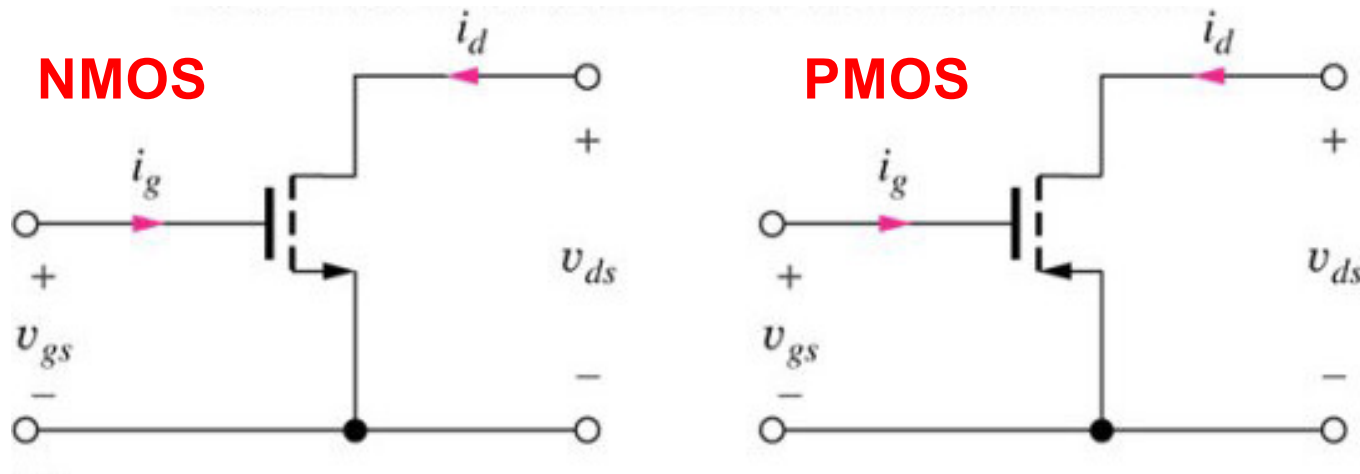
- By comparison we find

$$i_d = \frac{K_n}{2} [2(V_{GS} - V_T)v_{gs} + v_{gs}^2]$$

- For a linear relation between i_d and v_{gs} , we must request

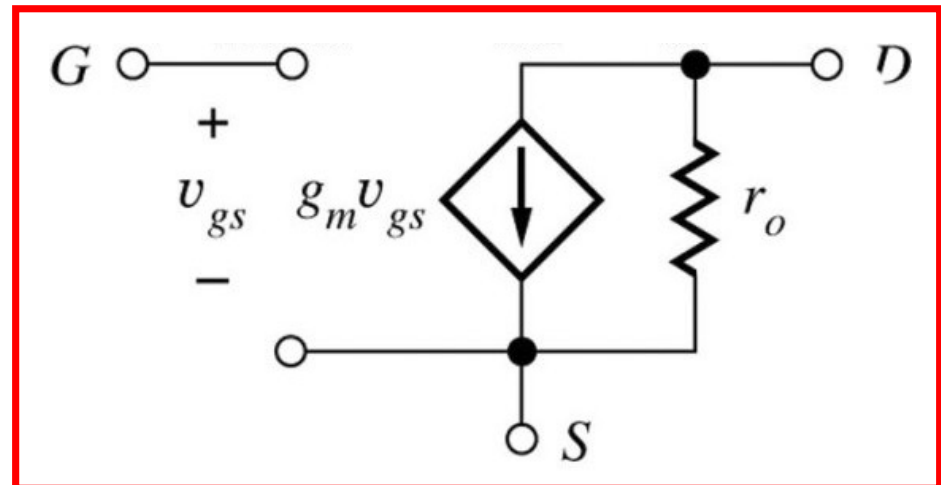
$$v_{gs}^2 \ll 2(V_{GS} - V_T)v_{gs} \quad \text{or} \quad v_{gs} \ll 2(V_{GS} - V_T)$$

Small-Signal Model for PMOS Transistors



- Small signal model for PMOS transistor is identical to that of the NMOS device

Jaeger, Blalock, Fig. 13.24



5.3.3 Comparison Small-Signal Model for BJT and MOSFET

Parameter	BJT	Relation	MOSFET
Transconductance g_m	$\frac{I_C}{(kT/q)}$	$\gg$	$\frac{2I_D}{V_{GS} - V_T} \cong \sqrt{2K_n I_D}$
Input Resistance r_π	$\frac{\beta_o}{g_m} = \frac{\beta_o (kT/q)}{I_C}$	$\ll$	∞
Output Resistance r_o	$\frac{V_A + V_{CE}}{I_C} \cong \frac{V_A}{I_C}$	$\approx$	$\frac{\lambda^{-1} + V_{DS}}{I_D} \cong \frac{1}{\lambda I_D}$
Amplification Factor $g_m r_o$	$\frac{V_A + V_{CE}}{(kT/q)} \cong \frac{V_A}{(kT/q)}$	$\gg$	$\frac{2(\lambda^{-1} + V_{DS})}{V_{GS} - V_T}$
Small-Signal Requirement	$v_{be} \ll (kT/q)$	$\ll$	$v_{gs} \ll 2(V_{GS} - V_T)$

$$I_C = I_S \left(e^{qV_{BE}/kT} - 1 \right) \left(1 + \frac{V_{CE}}{V_A} \right)$$

$$I_D = \frac{K_n}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

Comparison Small-Signal Model for BJT and MOSFET

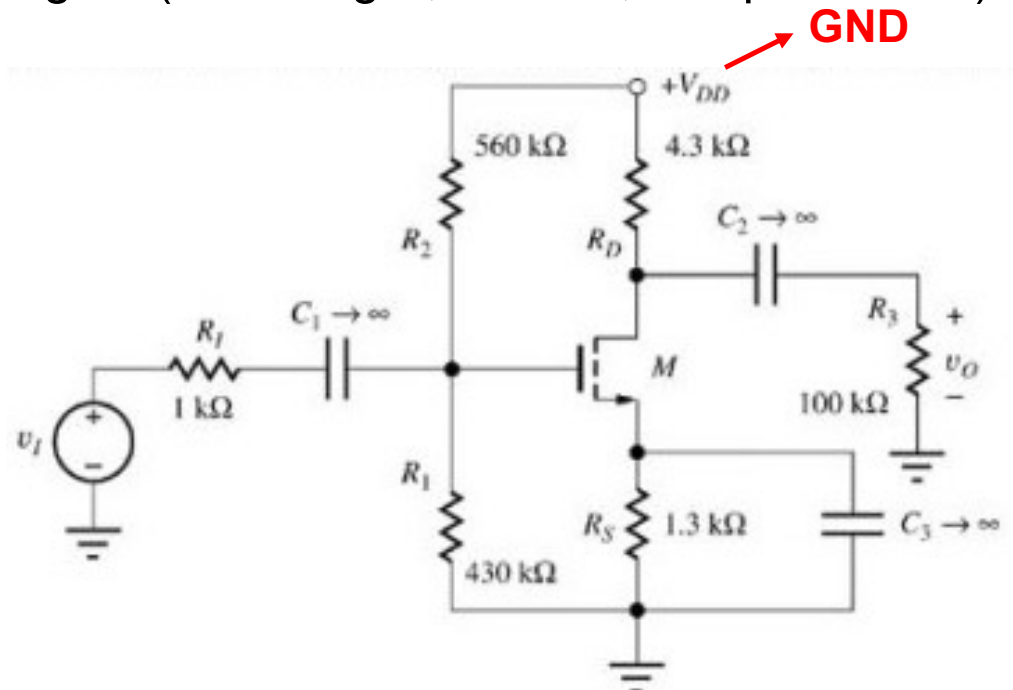
- Small signal parameters of BJT and MOSFET are directly controlled by the design of the Q-point
- For a given operating current, the MOSFET has a much smaller transconductance than the BJT
- The MOSFET transconductance increases with the square root of the drain current, the BJT transconductance is directly proportional to the collector current
- Transconductance of MOSFET depends on geometry W/L , transconductance of BJT is geometry independent
- Output resistance of BJT and MOSFET are similar for given operating point $(I_D, V_{DS}) = (I_C, V_{CE})$, because $1/\lambda$ is similar to V_A
- Amplification factor of BJT is larger than amplification factor of MOSFET
- Amplification factor of MOSFET decreases with increasing operating current, whereas amplification factor of BJT is independent of operating point
- MOSFET small signal model can handle larger values of v_{gs} than the v_{be} of the corresponding BJT

5.3.4 Common-Source Amplifier

- FET version of common-emitter amplifier (see Chapter 4.3)
- Goal: Calculate amplifier voltage gain (see Jaeger, Blalock, Chapter 13.10)

DC Analysis:

- Find equivalent circuit by replacing all capacitors with open circuits and inductors with short circuits
- Find the Q-point from the DC equivalent circuit using appropriate large-signal model for the transistor



AC Analysis:

- Find AC equivalent circuit by replacing all capacitors by short circuits and all inductors by open circuits, replacing DC voltage sources by ground connections and replacing DC current sources by open circuits
- Replace FET by its small-signal model
- Simplify the resulting AC circuit as much as possible

Common-Source Amplifier

– DC Analysis –

- Parameters: $R_1 = 430 \text{ k}\Omega$, $R_2 = 560 \text{ k}\Omega$, $R_D = 4.3 \text{ k}\Omega$, $R_S = 1.3 \text{ k}\Omega$, $V_{DD} = 12 \text{ V}$, $K_n = 0.5 \text{ mA/V}^2$, $V_T = 1 \text{ V}$, $\lambda = 0.0133 \text{ V}^{-1}$
- Thévenin equivalent of gate bias circuit:

$$V_{EQ} = \frac{R_1}{R_1 + R_2} V_{DD} = 5.21 \text{ V}$$

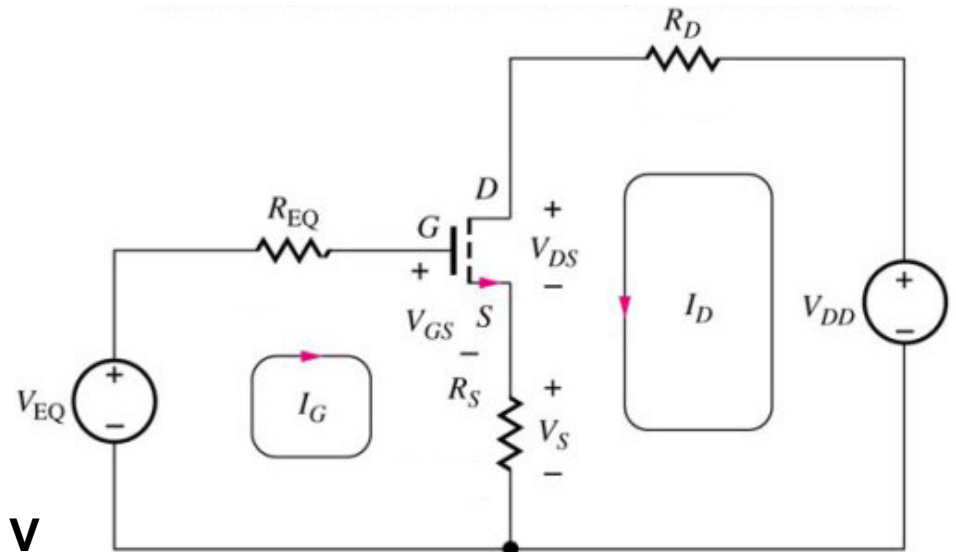
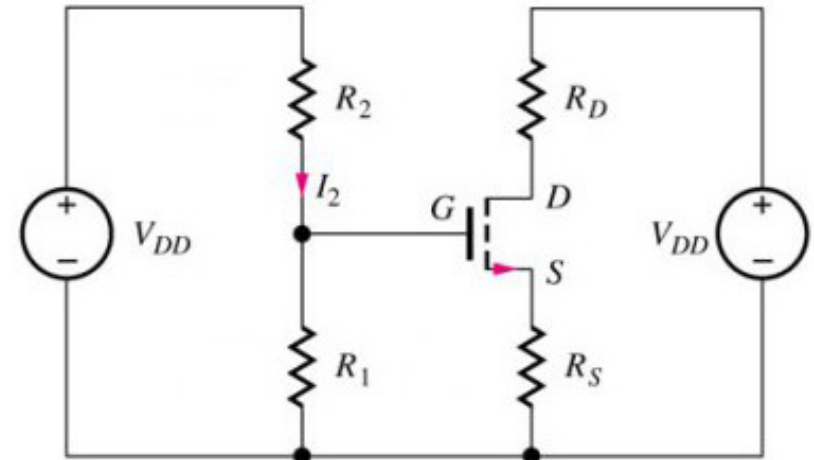
$$R_{EQ} = \frac{R_1 R_2}{R_1 + R_2} = 243.2 \text{ k}\Omega$$

- Extract V_{GS} , V_{DS} , I_D :

$$I_D = \frac{K_n}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

$$V_{EQ} = V_{GS} + I_D R_S$$

$$V_{DD} = (R_D + R_S) I_D + V_{DS}$$



→ $I_D = 1.44 \text{ mA}$, $V_{GS} = 3.34 \text{ V}$, $V_{DS} = 3.93 \text{ V}$

Common-Source Amplifier

– AC Analysis –

- Additional Parameters:
 $R_i = 1 \text{ k}\Omega$, $R_3 = 100 \text{ k}\Omega$
- $R_G = R_1 \parallel R_2 = 243.2 \text{ k}\Omega$
- Small signal parameters:

$$g_m = y_{21} = \frac{2I_D}{V_{GS} - V_T} = 1.23 \text{ mS}$$

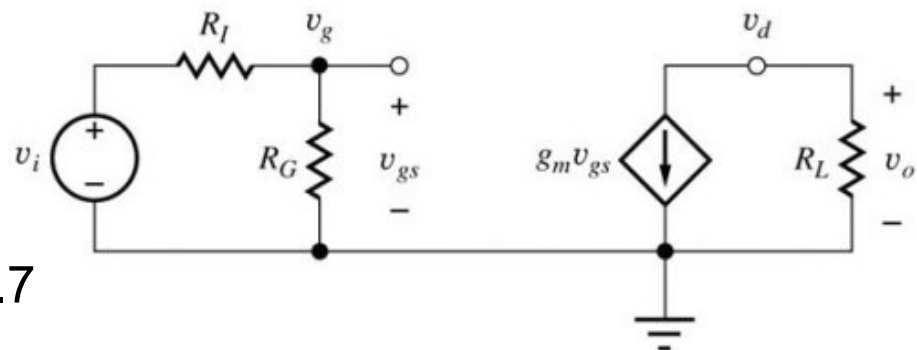
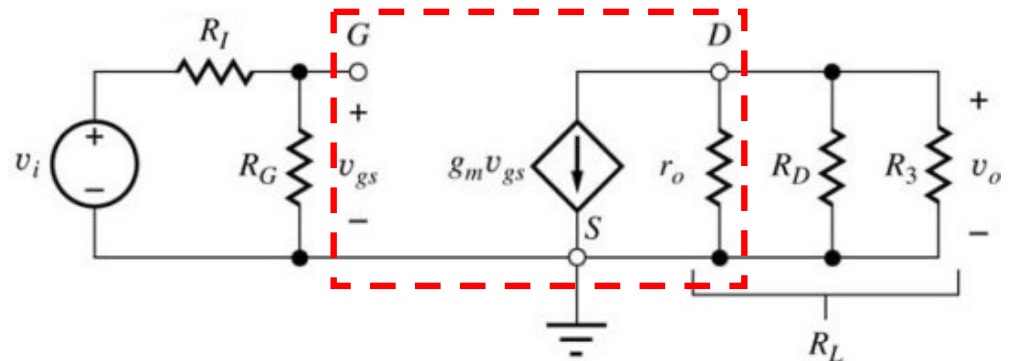
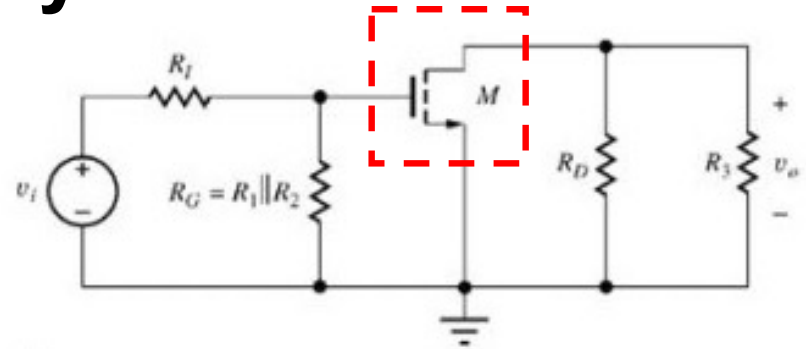
$$r_o = \frac{1}{y_{22}} = \frac{\lambda^{-1} + V_{DS}}{I_D} = 54.9 \text{ k}\Omega$$

- $R_L = r_o \parallel R_D \parallel R_3 = 3835 \text{ }\Omega$
- Calculate v_{gs} , v_o and v_o/v_i :

$$v_{gs} = \frac{R_G}{R_i + R_G} v_i = 0.996 v_i$$

$$v_o = -R_L g_m v_{gs} = -4.72 v_{gs}$$

$$A_v = v_o / v_i = -R_L g_m \frac{R_G}{R_i + R_G} = -4.7$$



Common-Source Amplifier

– AC Analysis –

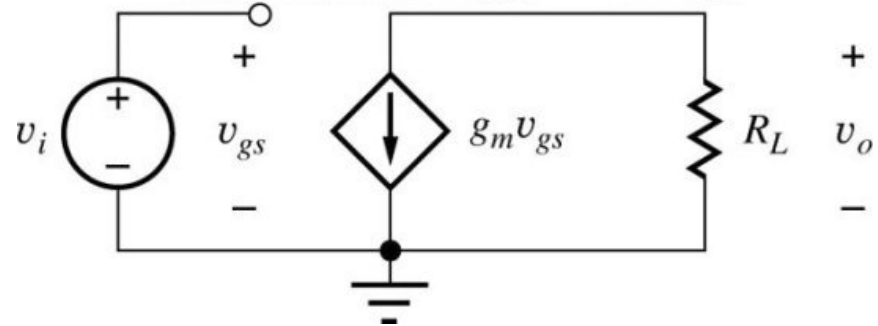
- If $R_G \gg R_i$, then $v_{gs} \approx v_i$ and the voltage gain of the common-source amplifier becomes

$$A_v \approx -g_m R_L$$

i.e. the product of the transistor transconductance g_m and the effective load resistor R_L

- The “–” sign indicates that input and output waveform are 180° out of phase
- The effective load resistance R_L consists of output resistance r_o in parallel with drain-bias resistor R_D and external load resistor R_3

Simplified equivalent circuit for $R_G \gg R_i$:



Common-Source Amplifier

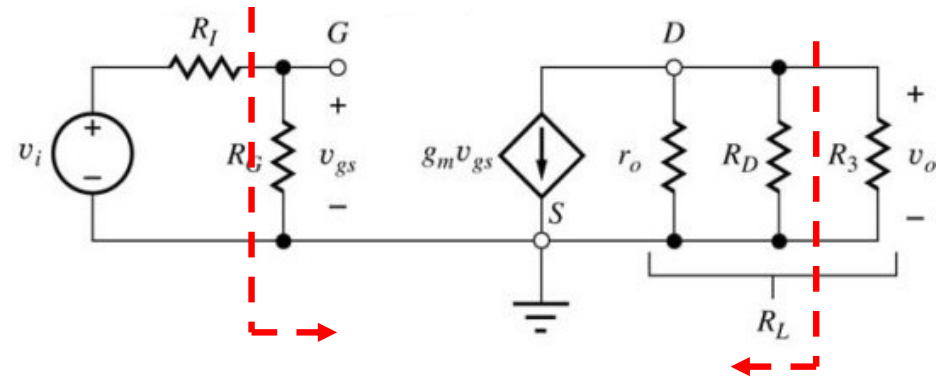
– Input/Output Resistance –

- **Input Resistance** = total resistance looking into amplifier at coupling capacitance C_1
- Attach test signal v_x to input side and measure i_x :

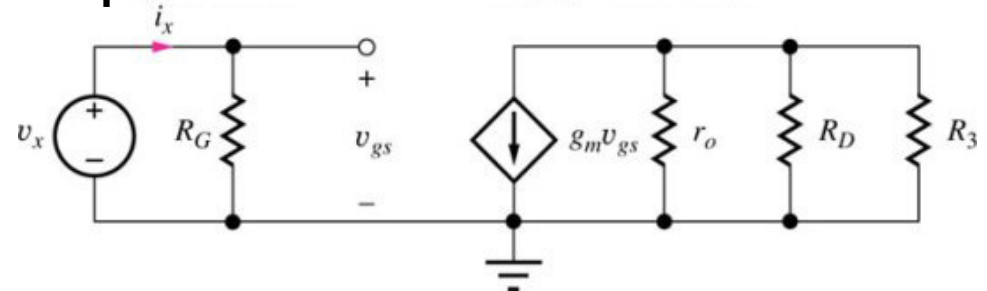
$$R_{in} = \frac{v_x}{i_x} = R_G = 243 \text{ k}\Omega$$

- **Output Resistance** = total resistance looking into amplifier at coupling capacitance C_2
- Attach test signal v_x to output side and measure i_x (keep in mind that $v_{gs} = 0$ in this case):

$$R_{out} = \frac{v_x}{i_x} = r_o \parallel R_D \stackrel{r_o \gg R_D}{\approx} R_D = 4.3 \text{ k}\Omega$$



Input Resistance



Output Resistance

